

PNEUMA TRAVEL REWARDS & PRODUCT ANALYTICS

Business & Product Analytics Case Study

Python • PostgreSQL • SQL • Power BI

Project Type Independent portfolio case study	Data Synthetic dataset
Focus Travel rewards, users, bookings & subscriptions	Author Piyush Mishra

Important: This report uses synthetic data created for learning and portfolio purposes. It does not use Pneuma's internal or confidential data.

1. Executive Summary

This project is a Pneuma-inspired travel rewards analytics case study. It models a user journey from signup and travel search through award results, clicks, booking attempts, completed bookings and subscriptions. The objective is to demonstrate how a Data Analyst can combine product, travel, loyalty and revenue data to identify measurable business opportunities.

The workflow combines Python/Pandas for data preparation and exploratory analysis, PostgreSQL for structured business questions, and Power BI for KPI-driven visualization and dashboarding.

Headline KPIs

Metric	Value	Metric	Value
Total Users	500	Total Searches	6,000
Total Bookings	578	Total Revenue	₹300,349
Miles Redeemed	46.23M	Taxes Paid	₹6.36M
Search → Booking	9.63%	Subscriptions	151

Portfolio outcome: The project demonstrates end-to-end analytics capability: data modeling, SQL questioning, KPI creation, product funnel analysis, business interpretation and recommendations.

2. Business Context & Problem Statement

Award-travel platforms help users discover ways to redeem credit-card points and airline miles for travel. A product in this space can generate connected datasets covering users, searches, award results, interactions, bookings, loyalty balances and subscriptions.

Business Problem

The analytical challenge is to understand where users are engaging, which travel products and routes attract demand, how users move through the search-to-booking funnel, and how loyalty activity connects with subscription revenue.

Key Business Questions

- Which routes and destinations are searched most often?
- Which airlines and cabin classes receive the most demand?
- Which devices are used for travel searches?
- What is the search-to-booking conversion?
- Which routes and airlines generate the most bookings?
- How many miles are redeemed and how much tax is paid?
- How is subscription revenue distributed between plans?
- What product or growth opportunities can be explored from these patterns?

3. Dataset & Data Model

Eight synthetic tables were created to represent the core product journey.

Table	Rows	Purpose
users	500	User profile, signup, city, type and age group
points_accounts	800	Loyalty program accounts and points balances
searches	6,000	Origin, destination, dates, cabin and device
award_results	6,910	Airline, loyalty program, miles, taxes and cash price
clicks	2,600	User interaction with award results
booking_attempts	950	Attempts made after result interaction
bookings	578	Completed booking records and redemption data
subscriptions	151	Monthly/annual subscription activity and revenue

Data Flow

Users → Searches → Award Results → Clicks → Booking Attempts → Bookings → Subscriptions

The tables are connected through user, search, result and interaction identifiers. This structure supports funnel analysis while keeping search, result, click and booking activity separately measurable.

4. Tools, Workflow & Analytical Method

Stage	Tool	Work Performed
Data preparation	Python / Pandas	Loaded CSVs; checked structure, nulls, duplicates and distributions
Exploration	Python / Pandas	Route, airline, cabin, device, user, booking and subscription analysis
Database analysis	PostgreSQL / SQL	Business questions, aggregations, joins and conversion metrics
Visualization	Power BI	Data model, DAX measures, KPI cards and interactive report pages
Documentation	GitHub	Project structure, SQL, screenshots, dashboard and report

Analytical Workflow

Synthetic Dataset

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Python / Pandas — cleaning & EDA

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PostgreSQL — SQL business analysis

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Power BI — data model, DAX & dashboard

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Business Insights

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Potential Recommendations

5. SQL Analysis

Twenty business questions were used to translate raw tables into decision-oriented metrics.

Question	Analysis
Q1	Top 10 most searched routes
Q2	Highest-search destinations
Q3	Highest-search origins
Q4	Cabin class demand
Q5	Searches by device
Q6	Award results by airline
Q7	Award results by loyalty program
Q8	Average miles required by airline
Q9	Average taxes by airline
Q10	Top 10 most active users
Q11	Search count by user
Q12	Search-to-booking conversion
Q13	Top booking routes
Q14	Bookings by airline
Q15	Total miles redeemed
Q16	Miles redeemed by airline
Q17	Total taxes paid
Q18	Subscription revenue by plan
Q19	Subscription rate by user type
Q20	Monthly search trend

Repository reference: The complete SQL script is stored in `sql/pneuma_analysis.sql`, including table creation, CSV imports, row-count checks and all 20 analysis queries.

6. Key Findings from the Synthetic Dataset

Area	Observation
Route demand	Delhi → Dubai (648), Delhi → London (614) and Mumbai → Dubai (608) are among the most searched routes.
Destination demand	Dubai (1,771), London (1,438) and Singapore (938) are the leading searched destinations.
Cabin demand	Economy has 3,070 searches, followed by Business with 2,433 and First with 497.
Device behavior	Mobile has 3,330 searches, compared with 2,280 on Desktop and 390 on Tablet.
Booking funnel	The Power BI dashboard reports a Search-to-Booking rate of 9.63% for the portfolio dataset.
Booking routes	Delhi → London (67), Delhi → Singapore (59) and Delhi → Dubai (56) are among the highest-booking routes.
Airline bookings	Lufthansa (82), Emirates (80) and Singapore Airlines (75) have the highest booking counts in the dashboard.
Miles redeemed	Singapore Airlines and Lufthansa are among the airlines with the highest total miles redeemed.
Subscriptions	There are 151 subscriptions; monthly subscribers are 101 and annual subscribers are 50.

Interpretation note: These findings describe the synthetic portfolio dataset. They should not be presented as Pneuma's actual internal customer, booking or revenue metrics.

7. Power BI Dashboard

The dashboard is organized into seven pages for executive, product, travel and loyalty analysis.

Dashboard Page	Purpose
Home	Navigation, project context and headline KPIs
Executive Overview	Users, searches, bookings, taxes, revenue, miles and search trend
Executive Overview 2	Top searched routes and destinations
Airlines Performance	Award results by airline, cabin demand and device usage
User Analytics	Active/repeat users, user types, cities and age groups
Travel Analytics	Booking routes, bookings by airline and cabin
Loyalty & Revenue	Miles redeemed, subscription revenue, subscribers and taxes by airline

Dashboard design focus

- Consistent navigation across analytical pages.
- KPI-first executive summary for quick scanning.
- Separate pages for users, travel, airlines and loyalty/revenue.
- DAX measures for reusable, relationship-aware metrics.

8. Potential Business & Product Recommendations

These recommendations are hypotheses generated from the synthetic analysis and should be validated with real product data.

Opportunity	Potential action
Improve the search-to-booking funnel	Analyze post-search drop-offs across award results, clicks and booking attempts to identify friction points
Prioritize high-demand routes	Monitor award discovery quality and recommendation relevance for frequently searched origin-destination pairs
Strengthen the mobile experience	Mobile represents the largest search device category in the dataset, so mobile search, filtering and booking flow should be optimized
Improve redemption-value transparency	Compare miles required, taxes and indicative cash ticket price to help users understand redemption value
Analyze subscription conversion	Track the path from free usage to PRO subscription and examine which behaviors are associated with conversion

Next analytical step: Replace synthetic data with production-level event data and add cohort retention, result-level availability quality, funnel timestamps, subscription renewal/churn and redemption-value metrics.

9. Skills Demonstrated & Conclusion

Skills Demonstrated

- Business problem understanding and KPI definition
- Python and Pandas for data loading and EDA
- PostgreSQL and SQL for aggregation and joins
- Power BI data modeling and DAX
- Dashboard design and information hierarchy
- Product funnel and travel-rewards analytics
- Business insight generation and recommendation framing
- GitHub project documentation and portfolio presentation

Conclusion

This case study demonstrates an end-to-end Data Analyst workflow: starting with a business context, building a structured synthetic dataset, validating and exploring the data, answering business questions with SQL, creating a Power BI dashboard, and translating analytical observations into potential product and growth opportunities.

The project is intentionally framed as a portfolio case study rather than an internal Pneuma project. Its purpose is to demonstrate analytical thinking, technical execution and the ability to connect data with business decisions.

Author

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Disclaimer: This independent portfolio project uses synthetic data inspired by publicly described award-travel business concepts. It does not contain or represent Pneuma's confidential, proprietary or internal data.